GUSTAVO A. CARDONA

Google Scholar | Webpage | LinkedIn
Newark, DE, USA
cardona@udel.edu | gacardonac2@gmail.com | +1 (321)-350-2497

RESEARCH INTERESTS

My research aims to enable specification-driven autonomy for multi-robot systems that interact with people. I automate synthesis and decision-making by integrating temporal logic–based formal methods, optimization, machine learning, and physically aware control. I explicitly account for uncertainty, scalability, and partial satisfaction when goals conflict or constraints are violated. I design algorithms for reliable human–robot interaction and multi-robot coordination that connect high-level specifications to low-level controllers, enabling the planning, explanation, and safe execution of complex missions. I validate these methods on real platforms, including quadrotors, ground robots, and humanoids.

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Mechanical and Nuclear Engineering — *Richmond, VA, USA*
Virginia Commonwealth University — (*Aug 2026-Present*)

EDUCATION

PhD, Lehigh University (Member: AIRLab) — *Bethlehem, PA, USA*
Ph.D. in Mechanical Engineering (GPA: 3.88/4.0) — *May'2025*
Thesis: Formal Methods for Multi-Robot Systems: Scalable and Robust Planning with Temporal Logic and Partial Satisfaction,
Award: 2025 Rossin College of Engineering and Applied Science Elizabeth V. Stout Dissertation Award,
Award: 2025 Outstanding Dissertation-Mechanical Engineering & Mechanics,
Advisor: *Dr. Cristian-Ioan Vasile*

Master, Universidad Nacional de Colombia (Member: DESYNC Lab) — Bogota, Colombia
Magister in Industrial Automation (GPA: 4.7/5.0) — *2023*
Thesis: Bioinspired Algorithm of Robot Network for Assistance in Search and Rescue Operations
Advisor: *Dr. Eduardo Mojica-Nava*

Bachelor, Universidad Nacional de Colombia (Member: DESYNC Lab) — Bogota, Colombia
Bachelor's in Electrical Engineering (GPA: 4.1/5.0, rank:3/33) — *2018*

RESEARCH EXPERIENCE

Postdoctoral Researcher, University of Delaware — *Newark, DE*
Host: *Dr. Panagiotis Artemiadis* — (*June 2025-Aug-2026*)
◦ Planning and coordination algorithms for physical human-humanoid interactions.

Research Intern, Mitsubishi Electric Research Laboratories (MERL) — *Cambridge, MA*
Host: *Dr. Stefano Di Cairano* — (*May 2024-Aug 2024*)
◦ Development and implementation of optimization-based algorithms for real-world applications of ground robot coordination, scheduling, and motion planning in uncertain surrounding environments.

Research Assistant, Explainable Robotics Lab (ERL), Lehigh University — *Bethlehem, PA*
Advisor: *Dr. Cristian-Ioan Vasile* — (*Aug 2020-May 2025*)
◦ Planning for multi-robot systems (modular, swarms, heterogeneous, and self-driving cars) using temporal logic formalisms to capture mission specifications. By modeling Mixed-Integer Linear Programming formulations that capture robot dynamics and specifications.

Volunteer Researcher, Swarms Lab, Lehigh University — *Bethlehem, PA*
Advisor: *Dr. David Saldaña* — (*Aug 2020-May 2025*)
◦ Design of the Catenary-robot, a cable propelled by two quadrotors in its ends. Control design for the

non-prehensile manipulation of objects without human intervention. Design of an adaptive controller to overcome dynamic uncertainties.

Volunteer Researcher, Robotics & Intelligent Systems Lab *Colombia*
Advisor: *Dr. Juan M. Calderon* (*2016-2020*)
∘ Coordination algorithms for swarms in search and rescue scenarios. Victim detection and classification using neural networks. Navigation learning in non-convex scenarios using Q-learning. Algorithms for swarms for the navigation in cluttered environments.

Research Assistant, DESYNC LAB, Universidad Nacional de Colombia *Colombia*
Advisor: *Dr. Eduardo Mojica-Nava* (*2016-2020*)
∘ Bioinspired algorithms for search and rescue scenarios. Cooperative transportation of objects using quadrotors. Exploration and navigation of environments using Voronoi cells and potential functions.

PUBLICATIONS

Journal Articles

[J9] **Cardona, G. A.***, Liang, K.*, Dai, H., & Vasile, C. I., An Iterative Approach for Heterogeneous Multi-Agent Route Planning with Resource Transportation and Temporal Logic Goals Under Uncertainty. (In preparation to IEEE-TRO)

[J8] **Cardona, G. A.**, Vasile, C. I., & Di Cairano. S., Truck Fleet Coordination for Warehouse Trailer Management by Temporal Logic with Energy Constraints. (In preparation to IEEE-TASE)

[J7] **Cardona, G. A.** and Vasile, C. I., Control Synthesis with Weighted Signal Temporal Logic under Partial Satisfiability Conditions. (In Preparation IEEE-TAC)

[J6] **Cardona, G. A.**, S. Kumbhar, & Artemiadis, P., Toward Seamless Physical Human-Humanoid Interaction: Insights from Control, Intent, and Modeling with a Vision for What Comes Next. Journal of Intelligent & Robotics Systems (2026)

[J5] **Cardona, G. A.**, Kamale, D., & Vasile, C. I. (2025). STL and wSTL control synthesis: A disjunction-centric mixed-integer linear programming approach. Nonlinear Analysis: Hybrid Systems, 56, 101576.

[J4] **Cardona, G. A.**, & Vasile, C. I. (2024). Planning for heterogeneous teams of robots with temporal logic, capability, and resource constraints. The International Journal of Robotics Research, 43(13), 2089-2111.

[J3] D'antonio, D. S., **Cardona, G. A.**, & Saldana, D. (2021). The catenary robot: Design and control of a cable propelled by two quadrotors. IEEE Robotics and Automation Letters, 6(2), 3857-3863, (RAL).

[J2] **Cardona, G. A.**, Ramirez-Rugeles, J., Mojica-Nava, E., & Calderon, J. M. (2021). Visual victim detection and quadrotor-swarm coordination control in search and rescue environment. International Journal of Electrical and Computer Engineering, 11(3), 2079.

[J1] **Cardona, G. A.**, & Calderon, J. M. (2019). Robot swarm navigation and victim detection using rendezvous consensus in search and rescue operations. Applied Sciences, 9(8), 1702.

Conference Articles

[C29] Brousseau, W.,* , **Cardona, G. A.***, Sadula, S., Vlachos, D. G., and Artemiadis, P., ChemTL: Tool-Aware Temporal Logic for the Autonomous Planning and Execution of Self-Driven Scientific Experiments (Submitted to ACC-2027)

[C28] Hensley, C., **Cardona, G. A.**, & Vasile, C. I.. Maximum Effort: A Methodology for Over-Satisfaction of Signal Temporal Logic Specifications for Motion Planning in Robotic Systems (Submitted to ACC-2027)

[C27] Karagulle, R., **Cardona, G. A.**, Ozay, N., & Vasile, C. I.. Expressive Power of WSTL Formulas for Learning-to-Rank. (Submitted to the 29th ACM International Conference on Hybrid Systems: Computation and Control (Accepted at CDC-2026).

[C26] **Cardona, G. A.**, Vasile, C. I., & Di Cairano, S. (2025, July). Truck Fleet Coordination for Warehouse Trailer Management by Temporal Logic with Energy Constraints. In 2025 American Control Conference (ACC) (pp. 4211-4217). IEEE.

[C25] **Cardona, G. A.***, Liang, K.*, & Vasile, C. I., An Iterative Approach for Heterogeneous Multi-Agent Route Planning with Resource Transportation Uncertainty and Temporal Logic Goals. In 2025 IEEE International Conference on Robotics and Automation (ICRA) (pp. 1-7). IEEE.

[C24] Liang, K., **Cardona, G. A.**, Kamale, D., & Vasile, C. I. (2024, December). Learning Optimal Signal Temporal Logic Decision Trees for Classification: A Max-Flow MILP Formulation. In 2024 IEEE 63rd Conference on Decision and Control (CDC) (pp. 8332-8337). IEEE.

[C23] Liang, K., **Cardona, G. A.**, & Vasile, C. I. (2024, May). An Iterative Approach for Heterogeneous Multi-Agent Route Planning with Temporal Logic Goals and Travel Duration Uncertainty. In 2024 IEEE International Conference on Robotics and Automation (ICRA) (pp. 257-263). IEEE.

[C22] **Cardona, G. A.**, and Vasile, C. I., Preferences on Partial Satisfaction using Weighted Signal Temporal Logic Specifications, In 2023 European Control Conference (ECC), Bucharest, Romania, 2023, pp. 1-6.

[C21] **Cardona, G. A.**, Kamale, D., & Vasile, C. I. (2023, May). Mixed Integer Linear Programming Approach for Control Synthesis with Weighted Signal Temporal Logic. In Proceedings of the 26th ACM International Conference on Hybrid Systems: Computation and Control (HSCC) (pp. 1-12).

[C20] **Cardona, G. A.**, Leahy, K., & Vasile, C. I. (2023, May). Temporal Logic Swarm Control with Splitting and Merging. In 2023 IEEE International Conference on Robotics and Automation (ICRA) (pp. 12423-12429). IEEE.

[C19] **Cardona, G. A.**, Saldaña, D., & Vasile, C. I. (2022, December). Planning for Modular Aerial Robotic Tools with Temporal Logic Constraints. In 2022 IEEE 61st Conference on Decision and Control (CDC) (pp. 2878-2883). IEEE.

[C18] **Cardona, G. A.**, & Vasile, C. I. (2022, December). Partial Satisfaction of Signal Temporal Logic Specifications for Coordination of Multi-robot Systems. In Algorithmic Foundations of Robotics XV: Proceedings of the Fifteenth Workshop on the Algorithmic Foundations of Robotics (pp. 223-238). Cham: Springer International Publishing.

[C17] **Cardona, G. A.**, D. S. D'Antonio, R. Fierro and D. Saldaña, Adaptive Control for Cooperative Aerial Transportation Using Catenary Robots, 2021 Aerial Robotic Systems Physically Interacting with the Environment (AIRPHARO), 2021, pp. 1-8.

[C16] **Cardona, G. A.**, D'Antonio, D. S., Vasile, C. I., & Saldaña, D. (2021, September). Non-prehensile manipulation of cuboid objects using a catenary robot. In 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 5270-5275). IEEE.

[C15] Pabon, J. D., **Cardona, G. A.**, Ospina, N. I., Calderon, J., & Mojica-Nava, E. (2021, September). Event-Triggered Control for Weight-Unbalanced Directed Robot Networks. In 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 5831-5836). IEEE.

[C14] **Cardona, G. A.**, D. Tellez-Castro, J. Calderon and E. Mojica-Nava, "Adaptive Multi-Quadrotor Control for Cooperative Transportation of a Cable-Suspended Load," 2021 European Control Conference (ECC), 2021, pp. 696-701, IEEE.

[C13] Paez, D., Romero, J. P., Noriega, B., **Cardona, G. A.**, & Calderon, J. M. (2021). Distributed Particle Swarm Optimization for Multi-Robot System in Search and Rescue Operations. IFAC-PapersOnLine, 54(4), 1-6.

[C12] **Cardona, G. A.**, Arevalo-Castiblanco, M., Tellez-Castro, D., Calderon, J., & Mojica-Nava, E. (2021, May). Robust adaptive synchronization of interconnected heterogeneous quadrotors transporting a cable-suspended load. In 2021 IEEE International Conference on Robotics and Automation (ICRA) (pp. 31-37). IEEE.

[C11] Arevalo-Castiblanco, M. F., Tellez-Castro, D., **Cardona, G. A.**, & Mojica-Nava, E. (2019, December). An adaptive optimal control modification with input uncertainty for unknown heterogeneous agents synchronization. In 2019 IEEE 58th Conference on Decision and Control (CDC) (pp. 8242-8247). IEEE.

[C10] **Cardona G. A.**, D. Yanguas-Rojas, M. F. Arevalo-Castiblanco and E. Mojica-Nava, "Ant-Based Multi-Robot Exploration in Non-Convex Space without Global-Connectivity Constraints," 2019 18th European Control Conference (ECC), 2019, pp. 2065-2070, IEEE.

[C9] **Cardona, G. A.**, Bravo, C., Quesada, W., Ruiz, D., Obeng, M., Wu, X., & Calderon, J. M. (2019, April). Autonomous navigation for exploration of unknown environments and collision avoidance in mobile robots using reinforcement learning. In 2019 SoutheastCon (pp. 1-7). IEEE.

[C8] **Cardona, G. A.**, Tellez-Castro, D., & Mojica-Nava, E. (2019). Cooperative transportation of a cable-suspended load by multiple quadrotors. IFAC-PapersOnLine, 52(20), 145-150.

[C7] León, J., **Cardona, G. A.**, Jaimes, L. G., Calderón, J. M., & Rodriguez, P. O. (2018, June). Rendezvous Consensus Algorithm Applied to the Location of Possible Victims in Disaster Zones. In International Conference on Artificial Intelligence and Soft Computing (pp. 700-710). Springer, Cham.

[C6] Quesada, W. O., Rodriguez, J. I., Murillo, J. C., **Cardona, G. A.**, Yanguas-Rojas, D., Jaimes, L. G., & Calderón, J. M. (2018, June). Leader-follower formation for uav robot swarm based on fuzzy logic theory. In International Conference on Artificial Intelligence and Soft Computing (pp. 740-751). Springer, Cham.

[C5] **Cardona, G. A.**, Yanguas-Rojas, D., Jaimes, L. G., Obeng, M., Wu, X., Calderon, J. M., & Castro, V. (2018, April). "Design of a Simplified Artificial Neuron (I&F) Model in FPGA". In SoutheastCon 2018 (pp. 1-6). IEEE.

[C4] Yanguas-Rojas, D., **Cardona, G. A.**, Ramirez-Rugeles, J., & Mojica-Nava, E. (2017, October). Victims search, identification, and evacuation with heterogeneous robot networks for search and rescue. In 2017 IEEE 3rd Colombian Conference on Automatic Control (CCAC) (pp. 1-6). IEEE.

[C3] León, J., **Cardona, G. A.**, Botello, A., Calderón, J.M. (2017). Robot Swarms Theory Applicable to Seek and Rescue Operation. In: Madureira, A., Abraham, A., Gamboa, D., Novais, P. Intelligent Systems Design and Applications. ISDA 2016. Advances in Intelligent Systems and Computing, vol 557. Springer, Cham

[C2] Calderon, J., **Cardona, G. A.**, Llofriu, M., Shamsi, M., Williams, F., Moreno, W., & Weitzenfeld, A. (2016, June). Impact force reduction using variable stiffness with an optimal approach for falling robots. In Robot World Cup (pp. 404-415). Springer, Cham.

[C1] **Cardona, G. A.**, Moreno, W., Weitzenfeld, A., & Calderon, J. M. "Reduction of impact force in falling robots using variable stiffness". In SoutheastCon 2016 (pp. 1-6). IEEE.

Workshop Articles and Preprints

[W2] **Cardona, G. A.**, Leahy, K., Mann, M., & Vasile, C. (2023). A Flexible and Efficient Temporal Logic Tool for Python: PyTeLo.

[W1] D'Antonio, D. S., Xu, J., **Cardona, G. A.**, & Saldaña, D. Customizable-ModQuad: a Versatile Hardware-Software Platform to Develop Lightweight and Low-cost Aerial Vehicles. (IROS-workshop 2022)

Software Tools Developed

[T1] A Flexible and Efficient Temporal Logic Tool for Python: PyTeLo.

INVITED PRESENTATIONS & DEMONSTRATIONS

Feb-2026 From High Level Planning and Physically towards Trustworthy Physical Human-Aware Control Robot Interaction, Florida State University, Tallahassee, FL, US.

Feb-2026 From High Level Planning and Physically towards Trustworthy Physical Human-Aware Control Robot Interaction, Montana State University, Bozeman, MT, US.

Jan-2026 From High Level Planning and Physically towards Trustworthy Physical Human-Aware Control Robot Interaction, Virginia Commonwealth University, Richmond, VA, US.

Jan-2026 From High Level Planning and Physically towards Trustworthy Physical Human-Aware Control Robot Interaction, University of Texas at El Paso, El Paso, TX, US.

May-2025 Formal Methods for Multi-Robot Systems: Scalable and Robust Planning with Temporal Logic and Partial Satisfaction, SAFE-Robotics Workshop, IEEE International Conference on Engineering Reliable Autonomous Systems (ERAS 2025), Worcester, MA, US.

Apr-2025 Formal Methods for Multi-Robot Systems: Scalable and Robust Planning with Temporal Logic and Partial Satisfaction, Safe Robotics Laboratory, Princeton University, US.

Mar-2025 Planning for Multi-robot Systems: A Formal Methods Approach, xLab, University of Pennsylvania, US.

Mar-2025 Planning for Multi-robot Systems: a Formal Methods Approach, Human-Oriented Robotics and Control Lab, University of Delaware, US.

Feb-2025 Planning for Muli-robot Systems: A Formal Methods Approach, Harvard Computational Robotics Group, Harvard University, US.

Jan-2025 Planning for Multi-robot Systems: A Formal Methods Approach, Walmart Advanced Systems and Robotics, US.

Aug-2024 Partial Satisfaction in Control Synthesis from Temporal Logic Specifications, SPARK Lab, Massachusetts Institute of Technology (MIT), US.

May-2024 Partial Satisfaction in Control Synthesis from Temporal Logic Specifications, Northeast Systems and Control Workshop, University of Pennsylvania (UPenn), US.

Oct-2021 Demonstration of LOMAP and TWTL at the workshop on Transforming Specifications into Robot Programs: A Survey of Formal Methods Tools for Non-Experts, *International Conference on Intelligent Robots and Systems (IROS)*.

Sept-2021 Geometric Controller for Quadrotors and Cooperative Transportation, Universidad Santo Tomas, Colombia.

MEMBERSHIP AND COMMUNITY SERVICE

- Mechanical Engineering Graduate Student Senate representative, Lehigh University, Fall 2024 - Spring 2025.
- Mechanical Engineering Graduate open house representative, Lehigh University, Fall 2022 - Fall 2024.

Membership

IEEE Member, IEEE RAS Member, IEEE CSS Member

Session Chair

- "Temporal Logic and Formal Verification", European Control Conference (ECC), Bucharest, Romania, 2023

Program Committee

- IEEE Colombian Caribbean Conference (C3), Barranquilla, Colombia, 2023

Workshop Organizer

- How to Ensure Correct Robot Behaviors? Software Challenges in Formal Methods for Robotics, In IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024.

Peer reviews

+85 Reviews verified on Web of Science (Clarivate)

Journals

- Automatica
- Autonomous Robots
- Robotics and Autonomous Systems
- IEEE Transactions on Robotics (TRO)
- IEEE Transactions on Automatic Control (TAC)
- International Journal of Robotics Research (IJRR)
- IEEE Transactions on Control Systems Technology (TCST)
- IEEE Transactions on Automation Science and Engineering (T-ASE)
- IEEE Robotics and Automation Letters (RA-L)
- Scientific Reports
- Journal of Control, Automation and Electrical Systems (JCAE)
- EURO Journal on Transportation and Logistics

Conferences

- International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Conference on Decision and Control (CDC)
- American Control Conference (ACC)
- European Control Conference (ECC)
- Learning for Dynamics & Control Conference (L4DC)
- IEEE International Conference on Control & Automation (ICCA)
- International Conference on Unmanned Aircraft Systems (ICUAS)
- International Conference on Advanced Robotics (ICAR)
- IEEE Colombian Conference on Automatic Control (CCAC)

- IEEE Colombian Caribean Conference (C3)

Event Organizer

- Volunteered Logistic Coordinator: Defend The Republic: Drone Competition at Lehigh University, 2023.

AWARDS AND ACCOMPLISHMENTS

- 2025 Rossin College of Engineering and Applied Science **Elizabeth V. Stout Dissertation Award**, Lehigh University
- **2025 Outstanding Dissertation-Mechanical Engineering & Mechanics**, Lehigh University
- ICRA 2023-**RAS Travel Grant**
- 2023 **Rossin Professional Development Program Fellowship**
- Rossin College of Engineering and Applied Science **RCEAS Fellowship**-Spring 2023
- **IEEE CSS Student Travel** Support Program of CDC 2022
- **Student Travel Grant WAFR**, The 15th International Workshop on the Algorithmic Foundations of Robotics (WAFR), 2022
- **Inclusion Fellow RSS**, Robotics Science and Systems Conference (RSS), 2022
- Graduate Student Senate **(GSS) travel grant** 2022
- **Fellowship**, Department of Mechanical Engineering and Mechanics in the Rossin Doctoral Fellows program for the academic year 2020
- **Academic merit award.** Teacher Assistant, exemption from payment of tuition fees. (Universidad Nacional de Colombia)
- **Automatic admission to graduate program**, best 10% graduated from bachelor's in Electrical Engineering (Universidad Nacional de Colombia)
- **Academic merit award.** Exemption from payment of tuition fees 2012 (Universidad Nacional de Colombia)

OUTREACH AND BROADENING PARTICIPATION IN STEM

- Lehigh University STEM council meeting to discuss challenges and improvement opportunities - 2024.
- Community outreach activities Lehigh Stem Squad: Rutgers Leap Academy, Drone competition. October 9, 2024
- PreLusion: Lehigh Women Engineers, robotics showcase for first-year students, Summer-2023.
- Robotics demonstration in the Choices Program, a project for middle school girls to get interested in STEM. Lehigh University, Summer-2023.
- Diversity, Equity, and Inclusion workshop (MEM-DEI), Seminar for the Mechanical Engineering and Mechanics Department, Lehigh University, Spring-2023 and Fall-2024.
- Workshop on Diversity, Equity, and Inclusion, AIRLab, Lehigh University, 2022.
- Proyecto SED-UN, teaching physics to 20 high-school students at "Colegio Republica Dominicana" to prepare for admission exam to the university. 2014.

TEACHING EXPERIENCE

- Teacher Development Program, Lehigh University, Level 1 and Level 2, 2023-2024.

Guest Lecture
◦ Digital Control (one lecture on observability to 15 undergraduate students), *Delaware University*, Fall 2025.
◦ Formal Methods in Robotics (4 lectures on MILPs and temporal logic to 10 graduate students), *Lehigh University*, Fall 2024.

Teaching Assistant *Lehigh University*
◦ Manufacturing (Laboratory Lecturer for 20 students) Spring 24
◦ Numerical Methods in Mechanical Engineering (Grader and office hours) Fall 22, Spring 23, Fall 23

Teaching Assistant (Lecturer) *Universidad Nacional de Colombia*
◦ Electric Machines. (Lecturer for more than 40 students) Spring 19, Fall 19

Undergraduate Teaching Assistant *Universidad Nacional de Colombia*
◦ Servo-Visual Control and Robotics (Tutoring for 20 students about OpenCV). Spring 18

Undergraduate Teaching Assistant *Universidad Nacional de Colombia*
◦ Electromagnetic Conversion Laboratory. (Lecturer for 20 students) Spring 17

Undergraduate Tutoring engineering program *Universidad Nacional de Colombia*
◦ Signals and Systems, Control Theory. (Tutoring for 60 students). Spring 17

ADVISING

Ph.D.
◦ David Paez, Virginia Commonwealth University, Richmond, VA, USA, Aug 2026-present.

Master
◦ Javier Alexis Cárdenas Bohorquez, Universidad Santo Tomas, Colombia, 2024-present (co-advised)
◦ Juan David Molina Rios, Universidad Santo Tomas, Colombia, 2025-present (co-advised)

MENTORING

PhD
◦ Hensley Crockett, Lehigh University, 2026-present
◦ Weston Brousseau, University of Delaware, 2026-present
◦ Shubham S. Kumbhar, University of Delaware, 2025-present
◦ Maria Alejandra Pinzon, Universidad Andres Bello, Chile, 2025-present
◦ Kaier Liang, Lehigh University, 2024-2026

Master
◦ Weston Brousseau, University of Delaware, 2025-2026
◦ Hengde Dai, Lehigh University, 2024-2025
◦ Thomas Snipes, Lehigh University, Summer-2024
◦ Juan David Pabon Hernandez, Universidad Nacional de Colombia, 2020

Undergraduate
◦ Nicolas Pollard, Virginia State University, VA, USA, Apr 2026 - present
◦ Brian Noriega, Universidad Santo Tomas, Colombia, 2021
◦ Juan P. Romero, Universidad Santo Tomas, Colombia, 2021
◦ David Paez, Universidad Santo Tomas, Colombia, 2021
◦ Juan Sebastian Ramirez Rugeles, Universidad Nacional de Colombia, 2019-2020

Technical Mentoring: Mountaintop Summer Experience Program, Mentoring more than 10 undergraduate students on developing a self-driving car platform (2022-2024).

IN THE NEWS

- Advancing autonomy, Resolve Magazine.
- MEM Celebrates Hispanic/Latinx Heritage Month
- Lehigh News: Advancing Autonomous Vehicles in Urban Spaces

SKILLS

Programming languages	Python, MATLAB
Environments	Linux, Windows
Simulation platforms	Gazebo, CoppeliaSim
Workflow and tools	Github, Jupyter Notebook, Matplotlib, PyTorch, OpenCV
Mathematical Skills	Numerical Methods, Optimization, Probability, Statistics
Other softwares	LaTeX, Arduino

REFERENCES

Available upon request.